

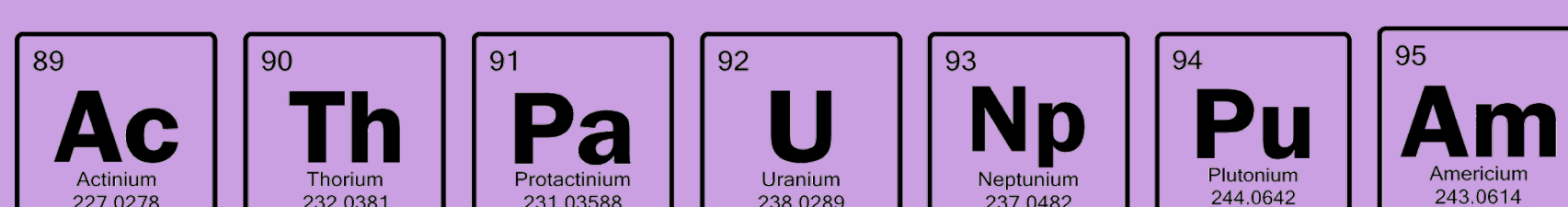
Computational f-Element Spectroscopy



Actinide Challenges

Understanding actinide electronic structure is essential for addressing important challenges in the nuclear fuel cycle:

wastewater remediation
environmental cleanup
long-term waste storage



Complex relativistic and multiconfigurational effects make actinide systems challenging to model and their spectra difficult to interpret from experiment alone.

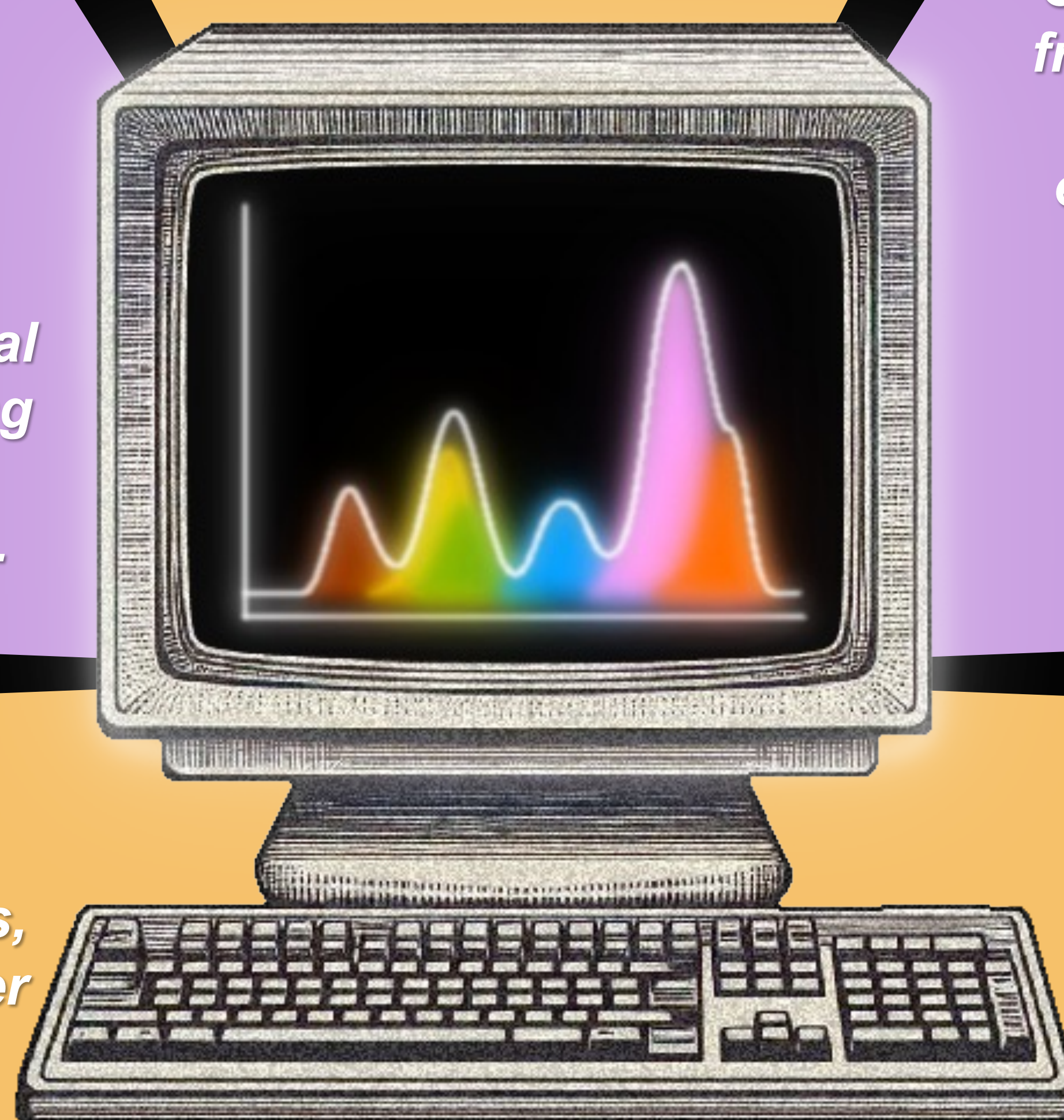
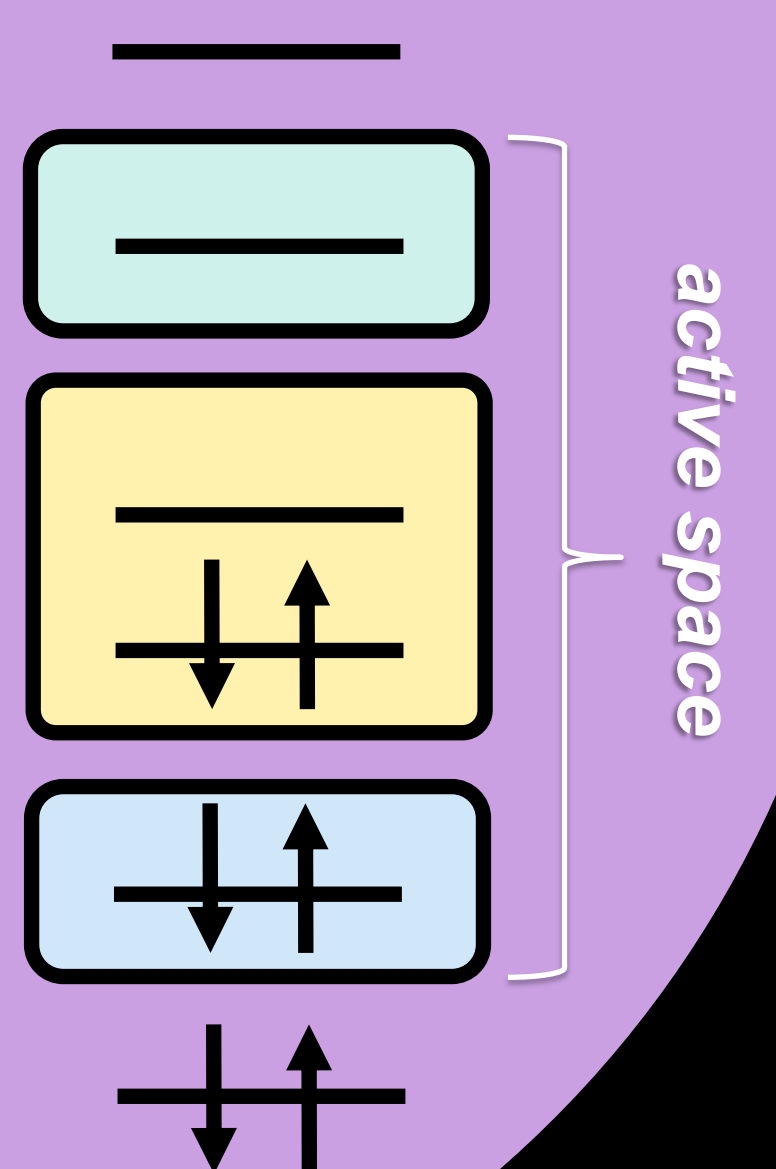
Computational Methods

We use complete active space self-consistent field (CASSCF-SO) methods to calculate the electronic structure and spectra of highly correlated f-element systems.

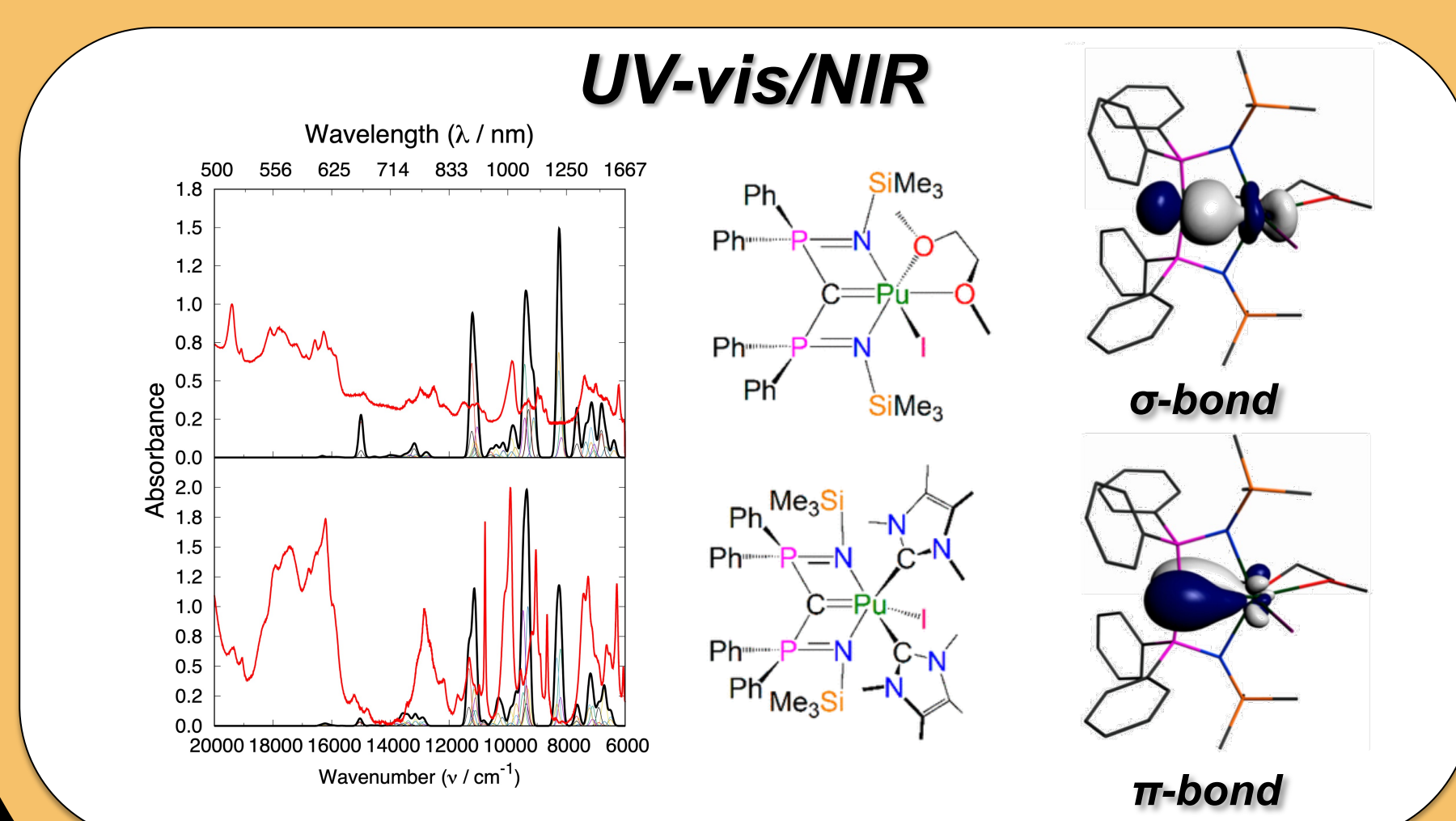
Our wavefunction is built from a linear combination of important electron configurations:

$$\Psi = \sum_i C_i \psi_i$$

JCTC, 2023, 19, 6933



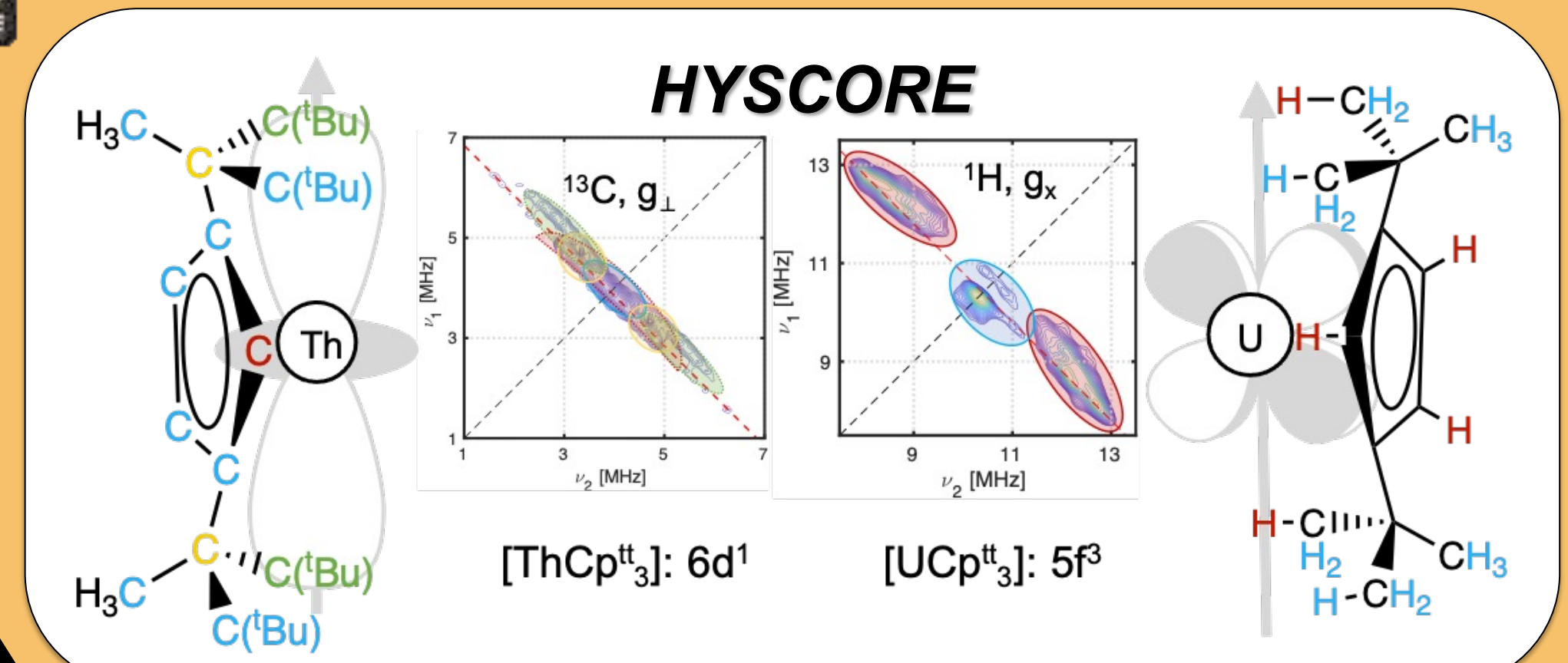
Correlating electronic structure with spectra reveals the nature of bonding interactions in actinide systems, aiding in development of new wastewater remediation techniques



Provides information about ligand-metal interactions

JACS, 2024, 146, 4098

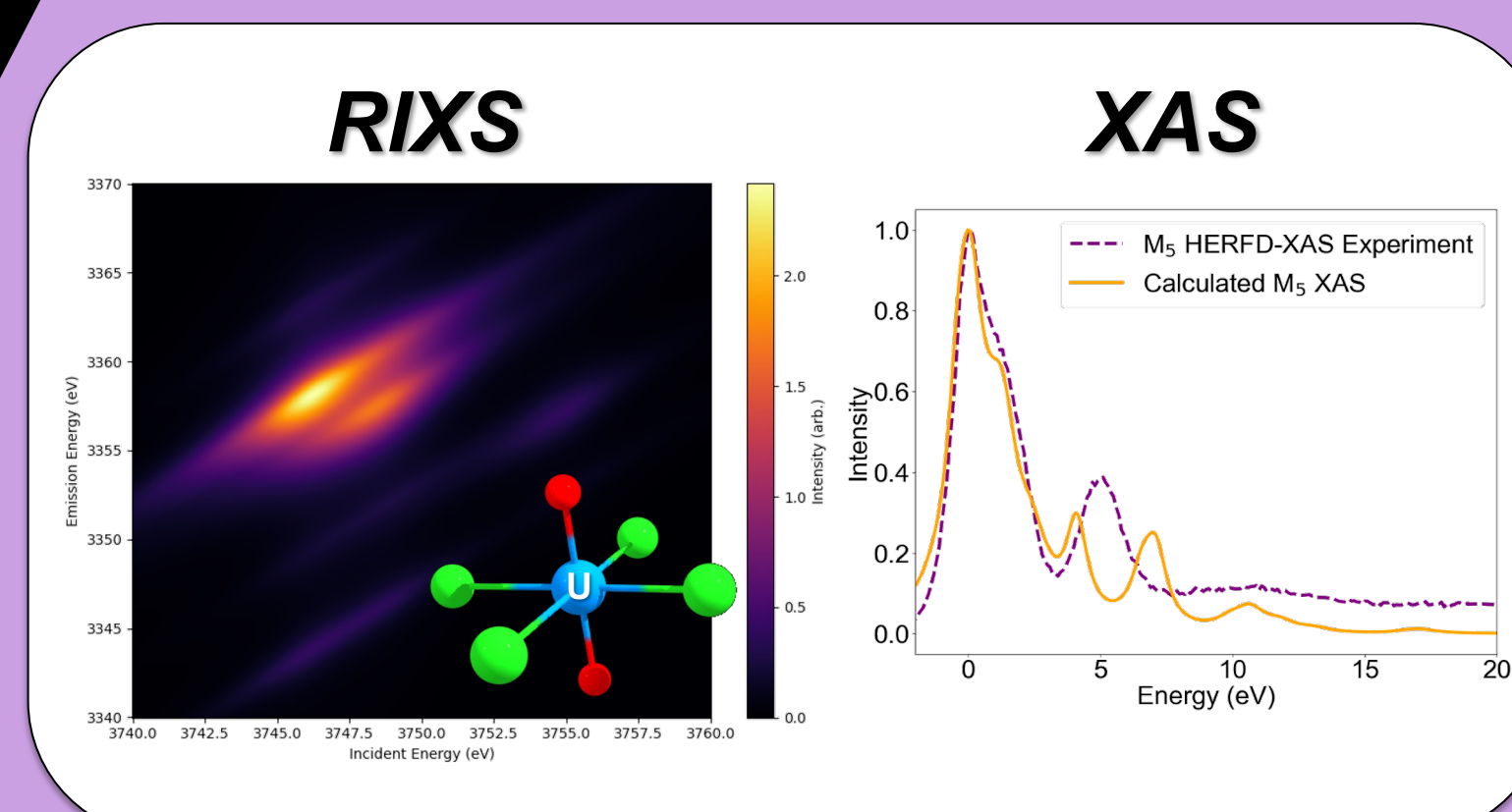
Deconvolution of HYSCORE spectra by computing relativistic hyperfine coupling constants and simulating EPR



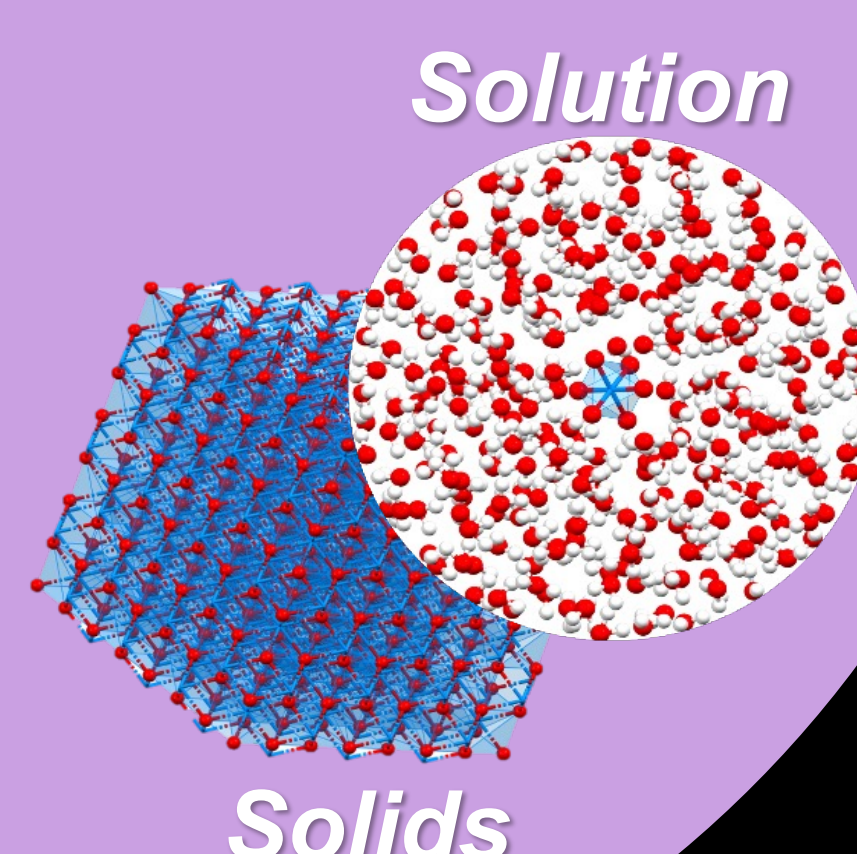
Probes covalency via unpaired ligand spin density

JACS, 2024, 146, 16660

Provides element-specific insight to actinide electronic structure (bonding, oxidation states)



We use cluster models to mimic realistic environments while keeping computational cost low



Magnetism

UV-vis/NIR

X-ray

